# Kuramoto SNN Object Recognition

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**Abstract**

## 1 Introduction

## 2 Method

Our model generates a separate mask for each discovered object by grouping image patches with similar temporal spike patterns. Inspired by S$^2$-Net (Dan and Wu 2026), it converts an image into patch activations and uses coupled oscillators to regulate when patches spike. Figure 1 shows the signal flow.

### 2.1 From Images to Neural Activations

**Patch activation.** We encode an RGB image with eight convolutional filters, normalize and bound the feature maps, and pool them into patch activations. The patch activation, $\gamma_i$, represents patch $i$. The patch indices, $i$ and $j$, identify image patches; the component index, $a$, identifies an oscillator component; and the time index, $t$, identifies a simulation step.

**Patch connection graph.** To connect related patches, we combine feature agreement with the paths between them.

$$S_{ij} = \frac{\boldsymbol{z}_i^\top \boldsymbol{z}_j}{\tau} - r\, d(i,j). \tag{1}$$

The connection score, $S_{ij}$, compares the normalized patch embeddings, $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$, using the temperature, $\tau$, and penalizes the geodesic distance, $d(i,j)$, using the spatial cost weight, $r$. $\tau$ and $r$ are derived from trainable scalars.

We normalize the strongest connections and make their weights reciprocal.

$$\begin{aligned} B_{ij} &= G\ \mathrm{softmax}_{j\in\mathcal{N}_i}(S_{ij}), \\ A_{ij} &= \frac{B_{ij} + B_{ji}}{2}. \end{aligned} \tag{2}$$

For each patch $i$, the neighbor set, $\mathcal{N}_i$, contains the 512 patches with the highest connection scores, $S_{ij}$. The connection gain, $G$, is derived from a trainable scalar. The normalized connection weight, $B_{ij}$, is zero for unselected neighbors. The symmetric connection weight, $A_{ij}$, defines the patch graph.

**Input phase.** A bounded phase input supplies image information to each oscillator.

$$\begin{aligned} \boldsymbol{u}_i &= W_{\mathrm{drive}}\gamma_i, \\ \boldsymbol{\phi}_i &= \pi \tanh(W_\phi\, \bar{\boldsymbol{u}}_i), \qquad \boldsymbol{\theta}_i(0) = \boldsymbol{\phi}_i. \end{aligned} \tag{3}$$

The projection matrix, $W_{\mathrm{drive}} \in \mathbb{R}^{4\times 8}$, and the phase gain, $W_\phi \in \mathbb{R}$, are trainable weights. The projected input, $\boldsymbol{u}_i$, is standardized to obtain $\bar{\boldsymbol{u}}_i$. The sensory drive, $\boldsymbol{\phi}_i$, initializes the oscillator phase, $\boldsymbol{\theta}_i(0)$.

### 2.2 Recurrent Core

**Kuramoto layer.** Oscillators coordinate their phases through intrinsic drift, patch interactions, and sensory drive.

$$\begin{aligned} \boldsymbol{\theta}_i(t+1) = \boldsymbol{\theta}_i(t) + \Delta t\Big(&\boldsymbol{\omega}_i + f\boldsymbol{\phi}_i \\ &+ \boldsymbol{\kappa}_i \odot \sin(\boldsymbol{\phi}_i - \boldsymbol{\theta}_i(t)) \\ &+ \frac{K}{N}\sum_j \bar{A}_{ij} \sin\big(\boldsymbol{\theta}_j(t) - \boldsymbol{\theta}_i(t) - \alpha_{ij}\big)\Big). \end{aligned} \tag{4}$$

The **natural frequency**, $\boldsymbol{\omega}_i$, gives each oscillator its intrinsic rate of phase change. The **coupling term**, the sum over $j$, adjusts the phase according to other patches, using the corrected connection weight, $\bar{A}_{ij}$, and the phase lag, $\alpha_{ij}$. The **drive term** uses the sensory drive, $\boldsymbol{\phi}_i$, to adjust frequency through $f$ and to pull the phase toward the input when the drive strength, $\boldsymbol{\kappa}_i$, is positive. The coupling scale, $K$, is normalized by the patch count, $N$. The time step, $\Delta t$, scales the update, and $\odot$ denotes elementwise multiplication. The parameters $\boldsymbol{\omega}_i$, $f$, and $\boldsymbol{\kappa}_i$ are trainable.

**Sinusoidal gating.** The patch rhythm controls how oscillatory signals enter the spiking core.

$$\begin{aligned} \bar{\theta}_i(t) &= \frac{1}{4}\sum_{a=1}^{4} \theta_{ia}(t), \\ g_i(t) &= \frac{1 + \sin(\bar{\theta}_i(\max(0, t-2)))}{2}. \end{aligned} \tag{5}$$

$$\boldsymbol{x}_i(t) = g_i(t)\sin(\boldsymbol{\theta}_i(t)). \tag{6}$$

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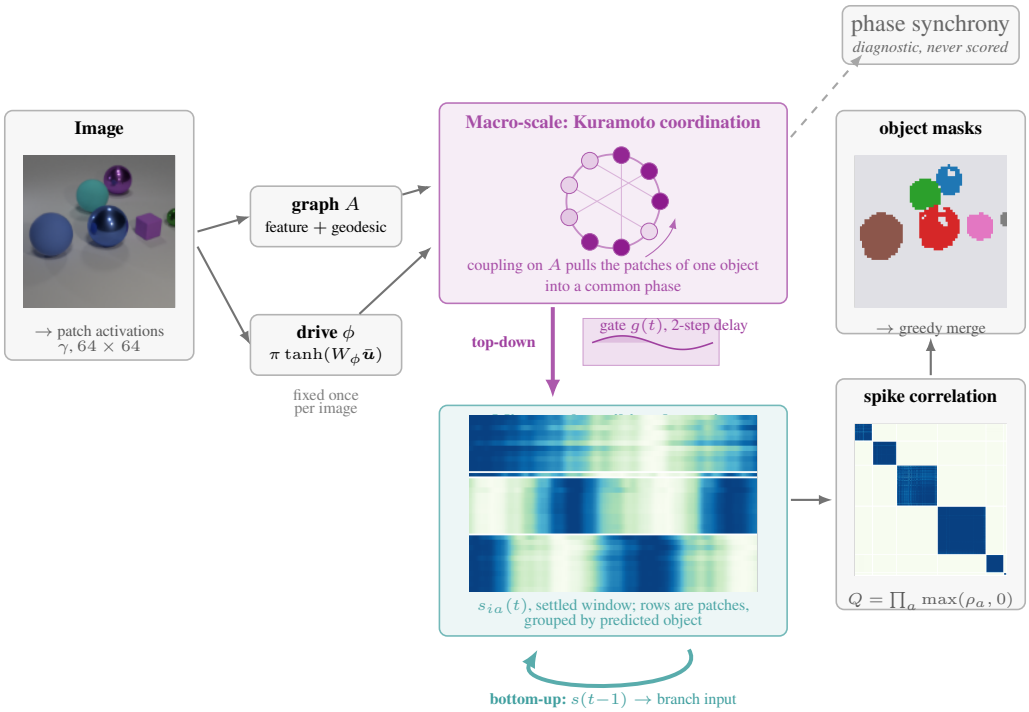



Figure 1: Signal flow, with the image, spiking, correlation and mask panels showing the trained model's actual output on one CLEVR image from the reserved split (`image_id` 1355, six objects; FG-ARI 0.9158, the median of the 121 reserved scenes with four to six objects, against a split mean of 0.8421). The mask shown is bit-identical to the one that was scored. An image is encoded into patch activations $\gamma$, which fix both the coupling graph $A$ and the sensory drive $\phi$ once per image. Oscillator phases $\theta$ evolve under Kuramoto coupling on $A$ (ring, illustrative). The mean patch phase gates the spiking layer two steps later (**top-down**, violet), and within that layer the previous spike re-enters the dendritic branch input (**bottom-up**, teal). In the spiking panel, patches the readout assigns to one object share a band pattern and different objects are offset; the correlation panel is that structure as the affinity $Q$, ordered by predicted object, and its blocks are the groups that become masks. The mask panel has the minimum-region-size rule applied ($f = 0.01$, fixed on validation): without it this image yields eleven regions of sizes $285, 240, 141, 106, 102, 16, 1, 1, 1, 1, 1$, and the rule removes the five single-patch ones, leaving six for six true objects.

The mean patch phase, $\bar{\theta}_i(t)$, summarizes the four oscillator phases in a patch. The delayed gate, $g_i(t)$, controls signal transmission using the patch phase from two steps earlier. The gated phase feature, $\boldsymbol{x}_i(t)$, carries the modulated oscillations to the dendritic layer.

**Dendritic layer.** Four branches process the rhythmic input and previous spike before combining their responses.

$$\boldsymbol{v}_{ia}(t) = U \begin{pmatrix} x_{ia}(t) \\ s_{ia}(t-1) \end{pmatrix} + \boldsymbol{b}. \tag{7}$$

The branch input, $\boldsymbol{v}_{ia}(t)$, combines the gated feature, $x_{ia}(t)$, and the previous emitted spike, $s_{ia}(t-1)$. The previous spike provides top-down feedback from the spiking output to the dendritic branches. The projection weight, $U \in \mathbb{R}^{4\times 2}$, and the bias, $\boldsymbol{b} \in \mathbb{R}^4$, are trainable.

The provisional branch state, $\widehat{\boldsymbol{h}}_{ia}(t)$, balances the previous state and the current branch input through the retention vector, $\boldsymbol{\beta}_i$:

$$\begin{aligned} \widehat{\boldsymbol{h}}_{ia}(t) = \boldsymbol{\beta}_i \odot \boldsymbol{h}_{ia}(t-1) \\ + (1 - \boldsymbol{\beta}_i) \odot \boldsymbol{v}_{ia}(t). \end{aligned} \tag{8}$$

$$\begin{aligned} \boldsymbol{h}_{ia}(t) &= \boldsymbol{h}_{ia}(t-1) + \big(1 - \eta_a(1 - g_i(t))\big) \\ &\quad \cdot \big(\widehat{\boldsymbol{h}}_{ia}(t) - \boldsymbol{h}_{ia}(t-1)\big), \\ H_{ia}(t) &= \sum_{b=1}^{4} h_{iab}(t). \end{aligned} \tag{9}$$

The branch state, $\boldsymbol{h}_{ia}(t)$, moves toward the provisional branch state, $\widehat{\boldsymbol{h}}_{ia}(t)$, according to the gate and the trainable modulation weight, $\eta_a$. The branch index, $b$, selects an entry of $\boldsymbol{h}_{ia}(t)$; the membrane input, $H_{ia}(t)$, sums all four entries.

**Membrane layer.** The provisional membrane voltage, $\widehat{V}_{ia}(t)$, balances the previous voltage and the current membrane input through the retention coefficient, $\lambda_i$, then subtracts a reset proportional to the previous spike:

$$\begin{aligned} \widehat{V}_{ia}(t) = \lambda_i V_{ia}(t-1) + (1 - \lambda_i) H_{ia}(t) \\ - \vartheta_a s_{ia}(t-1). \end{aligned} \tag{10}$$

$$\begin{aligned} V_{ia}(t) = V_{ia}(t-1) + \big(1 - \xi_a(1 - g_i(t))\big) \\ \cdot \big(\widehat{V}_{ia}(t) - V_{ia}(t-1)\big). \end{aligned} \tag{11}$$

$$e_{ia}(t) = \mathrm{ind}(V_{ia}(t) > \vartheta_a),$$
$$s_{ia}(t) = g_i(t) e_{ia}(t). \tag{12}$$

The membrane voltage, $V_{ia}(t)$, moves toward the provisional voltage, $\widehat{V}_{ia}(t)$, according to the gate and the trainable modulation weight, $\xi_a$. The firing threshold, $\vartheta_a$, is derived from a trainable parameter. The indicator function, $\mathrm{ind}$, produces the binary firing event, $e_{ia}(t)$, which the gate scales to produce the emitted spike, $s_{ia}(t)$.

### 2.3 Spike Grouping and Mask Generation

We mean-center and L2-normalize the final 512 emitted spike samples for each oscillator component. The component correlation, $\rho_{ija}$, sums the products of corresponding samples in the normalized traces:

$$\rho_{ija} = \mathrm{clip}\left(\sum_{m=1}^{512} r_{ia,m}\, r_{ja,m}, -1, 1\right), \tag{13}$$

where the sample index, $m$, runs over the retained time window. To obtain the patch affinity, $Q_{ij}$, we replace negative correlations with zero and multiply the resulting values across the four components. We form connected components by linking patch pairs with $Q_{ij} \geq 0.5$, discard components smaller than 8 patches, a floor selected on CLEVR validation, and treat the largest remaining component as background. Every patch enters this step as a candidate; no firing-rate test is applied, so the foreground is defined entirely by the synchrony structure.

We start the retained foreground patches as singleton groups and merge spatially adjacent groups using

$$E_{ij} = \left(1 + \frac{N\bar{A}_{ij}}{G}\right) \log \frac{Q'_{ij}}{1 - Q'_{ij}},$$
$$\Delta(\mathcal{C}, \mathcal{D}) = \sum_{i \in \mathcal{C}} \sum_{j \in \mathcal{D}} E_{ij}. \tag{14}$$

The stabilized affinity, $Q'_{ij}$, determines the sign of the merge evidence, $E_{ij}$. The merge score, $\Delta(\mathcal{C}, \mathcal{D})$, sums the evidence between the candidate patch groups, $\mathcal{C}$ and $\mathcal{D}$. Groups are merged while their score is positive. Each final group produces an object mask.

## 3 Experiments

### 3.1 Protocol

All models are evaluated by one function on one representation. Ground truth is quantised to a $64 \times 64$ patch grid and scored there; a prediction is *never* upsampled to pixels to be compared. A prediction coarser than the grid is lifted by an exact `repeat`, and one finer than the grid is reduced by patch majority vote, so every method is scored on the same object. The ceiling is therefore $1.0$ by construction, and these numbers are not directly comparable with pixel-protocol figures reported elsewhere.

We report three metrics and never combine them into a single score: foreground adjusted Rand index (FG-ARI), foreground IoU (FG-IOU) and matched-object IoU (OBJ-IOU). Background is label $0$ on both sides. They differ in which patches they look at: FG-ARI restricts both label maps to the patches whose *ground truth* is foreground and scores the two partitions of that set, so a patch the truth calls background is removed before scoring and predicting an object there is invisible to it, while FG-IOU merges every non-background identity on each side and compares the two sets over the union of predicted and true foreground, which makes it the metric that charges for that error. Appendix E gives exact definitions.

**Matched exposure.** Every model is trained on the *same* $10{,}000$ CLEVR images for $10$ epochs. For our model that is $6{,}250$ updates at batch $16$, i.e. $100{,}000$ image exposures; a baseline whose batch size does not divide $10{,}000$ sees a few more through its loader's rounding. Where a baseline's published recipe differs, only its epoch count is changed; architecture and optimiser are left at the authors' values. Tetrominoes and Multi-dSprites are never trained on: for every row in Table 1 those two columns are zero-shot transfer from the CLEVR-trained model. Training on the target and tuning on the target are separate conditions and do not hold equally for every row; Appendix E states which split fixed each model's free choices.

**Readouts that require a cluster count.** Several baselines produce a continuous embedding and obtain masks by clustering it. We found that the reference implementations of CAE, CAE++, CtCAE and SynCx set the number of clusters from the *per-image ground-truth object count*,

$$k_i \;=\; \big|\mathrm{unique}(y_i)\big|,$$

which makes their published figures oracle-assisted. For the comparison to be meaningful we removed this: $k$ is a per-dataset constant selected on that dataset's own validation split, never read from the evaluated image. Cluster identities are arbitrary, so the mapping of one cluster to background is likewise fixed by a rule chosen on the same validation split. That is a weaker transfer condition than ours, where the one free scalar is fixed on CLEVR validation and transferred unchanged. No hyperparameter, threshold or rule of any model is selected on the reported split.

**Minimum region size.** Our readout emits a sizeable number of single-patch regions: $22.8\%$ of all predicted regions on CLEVR validation, $27.0\%$ on Tetrominoes and $56.2\%$ on Multi-dSprites, where the *median* predicted region is one patch. They arise after the synchrony grouping, when the greedy merge splits a spatially scattered synchronous component into pieces, and they inflate the object count without materially affecting the partition. We therefore discard any predicted region smaller than a fraction $f$ of that image's predicted foreground, reassigning it to background. The rule is scale-free, reads no ground truth, and $f$ is a single constant fixed on CLEVR validation as the largest value whose FG-ARI loss stays under $0.005$ and whose matched-object IoU loss stays under $0.01$, giving $f = 0.01$. The same $f$ is applied unchanged to Tetrominoes and Multi-dSprites — no target-domain data is used to set it — and to every baseline.

### 3.2 Generalisation across datasets

Table 1 is the central result. Every model sees only CLEVR during training; the Tetrominoes and Multi-dSprites columns measure transfer.

Under the matched training budget, our model scored higher on all nine dataset–metric combinations, at a single training seed. The reference row takes the *best baseline per cell* separately, so it is a composite of three different architectures that no single baseline attains. *Which* baseline is second depends on the dataset — AKOrN (conv) on Tetrominoes, AKOrN (attn)+up-tiling on Multi-dSprites, ItrSA on CLEVR matched-object IoU.

**Where the margin is thin.** The smallest margin in the table is Tetrominoes foreground IoU, $+0.0410$. Every other margin lies between $0.128$ and $0.326$. We flag this cell explicitly rather than quoting only the aggregate.

**Predicted foreground.** A single diagnostic separates grouping from tiling. Against a true foreground fraction of $0.2115$ on CLEVR and $0.0943$ on Multi-dSprites, our readout predicts $0.1988$ and $0.0827$ — the only method that does not over-claim — while baselines predict between $0.41$ and $0.86$ on CLEVR. A model that labels most of the image as object can score well on a foreground-only partition measure while failing the IoU metrics, which is what the F-IOU column shows.

### 3.3 Budget and released weights

Two conditions on Table 1 deserve measurements rather than argument: that the baselines are evaluated away from their published setting, and that the matched budget limits them.

**Released weights.** Only Slot Attention ships trained weights; the AKOrN, CAE and SynCx repositories release data but no checkpoints. Evaluated directly, at roughly $320\times$ our exposure budget, its released model scores $.3747/.2219/.1165$ on CLEVR, $.4742/.2381/.2355$ on Tetrominoes and $.3205/.0980/.1204$ on Multi-dSprites. Our model scores higher in all nine cells, by $0.377$ to $0.720$, so the comparison does not rest on our retraining of it. The extra data also helps it far more *off* its training distribution than on it — worth $+.0875$ FG-ARI on CLEVR but $+.3003$ on Tetrominoes and $+.1275$ on Multi-dSprites — so a data-efficiency conclusion drawn from an in-domain column alone would not transfer.

**Budget.** Comparing each baseline against its own published number, by its own evaluation code, isolates the effect of the shortened schedule. Two of the six are *not* budget-limited: AKOrN (conv) reaches $60.5$ at $1/210$ of its schedule against a published $59.0 \pm 4.3$, and CAE reaches $10.3$ at $1/64$ against a published $4.0$. For those two the margin in Table 1 is not an artefact of a short schedule. The other four are limited — AKOrN (attn) $60.4$ against $91.0 \pm 0.5$, ItrSA $58.1$ against $82.5 \pm 1.7$, CAE++ $16.7$ against $27.0$, and Ct-CAE $12.0$ against $54.0$ — and against those the result should be read as a data-efficiency claim. We state both readings rather than only the favourable one. Full figures are in the supplementary document.

### 3.4 Rollout length

The readout integrates spiking activity over $T$ simulation steps and reads the second half of the window. Table 2 varies $T$ with the model fixed.

Halving the rollout costs $0.0019$ FG-ARI — below any difference we can resolve at one seed — and *improves* foreground IoU. Performance degrades gently to $T = 256$ and collapses at $T = 128$, where the model also under-segments, predicting a foreground fraction of $0.1148$ against a truth of $0.2115$. We therefore use $T = 512$ as the default operating point and $T = 1024$ for the reported numbers. Shorter windows also fragment more: mean predicted regions per image rise from $8.0$ at $T = 1024$ to $13.3$ at $T = 256$, against $6.36$ true objects.

**Scaling with training data.** Continuing training on additional unseen CLEVR images, in steps of $1{,}000$ from $10{,}000$ to $18{,}000$, separates the three metrics. FG-ARI runs $.8421$, $.8426$, $.8449$, $.8460$, $.8469$, $.8459$, $.8456$, $.8470$, $.8461$: it gains $.0028$ over the first two steps and then stays inside a band of $.0021$ for the remaining six, with no trend. Foreground IoU runs $.7545$ to $.7755$ and matched-object IoU $.6410$ to $.6579$, both still rising at $18{,}000$ and neither turning. More in-domain data therefore stops improving *how* patches are grouped at around $12{,}000$ images while it is still improving *where the boundaries fall*. Mean predicted regions hold at $7.7$–$8.0$ throughout, against $6.36$ true objects, so the extra data sharpens the partition without changing how much it splits. The FG-ARI band is one seed wide and we read it as flat rather than converged; the two IoU curves have not been run far enough to find their own ceiling.

**In-domain data is not the most useful data.** The same $5{,}000$ additional images buy more than twice as much when they come from a different dataset. Continuing on CLEVR raises CLEVR FG-ARI by $+.0038$; continuing on Tetrominoes instead raises it by $+.0084$, a $2.2\times$ larger gain at an identical budget. Together with the transfer result in Table 1, this argues that the model is not fitting a domain but learning a grouping rule that the extra renders refine.

### 3.5 Oscillatory timing, measured

The three measurements summarised in Section 3.2, in full. All run on the first $40$ images of the CLEVR reserved split, on the same rollouts, so they are directly comparable. They are reported in the order they answer the hypothesis: which pattern is observed, whether its temporal alignment is necessary, and which signal carries it.

**Validity of the ablation.** The affinity $Q$ is recomputed here from `component_spikes` rather than taken from the model, so that a modified spike trace can be fed through the same arithmetic. On the unshifted control the recomputed $Q$ agrees with the model's own to $6.7 \times 10^{-6}$ maximum absolute difference over all $4096^2$ entries, so the ablations differ from the control in exactly one thing.

**Separability.** Ground-truth labels are used only to sort patch pairs into three groups and never to set a threshold or a count. Averaged over the $40$ images, the affinity is

| Model | CLEVR (in-domain) | | | Tetrominoes (zero-shot) | | | Multi-dSprites (zero-shot) | | |
|---|---|---|---|---|---|---|---|---|---|
| | ARI | F-IOU | O-IOU | ARI | F-IOU | O-IOU | ARI | F-IOU | O-IOU |
| **Ours** | **.8408** | **.7528** | **.6387** | **.8568** | **.7950** | **.8170** | **.6966** | **.8190** | **.7625** |
| AKOrN (attn) + up-tiling | .6326 | .3468 | .3636 | .5484 | .7230 | .4167 | .5505 | .6539 | .6348 |
| AKOrN (conv) | .6046 | .4219 | .2769 | .7202 | .7540 | .5913 | .4055 | .6834 | .5434 |
| AKOrN (attn) | .6027 | .3327 | .3750 | .5676 | .7383 | .4687 | .4343 | .6019 | .5751 |
| ItrSA | .5794 | .2787 | .3853 | .6319 | .7137 | .5687 | .4817 | .2489 | .6093 |
| CAE++ | .2838 | .2193 | .0882 | .2715 | .2452 | .1901 | .2648 | .1048 | .1125 |
| Slot Attention | .2872 | .2220 | .0801 | .1739 | .2192 | .1071 | .1930 | .0876 | .0680 |
| CtCAE | .2357 | .2148 | .0964 | .2681 | .2420 | .2034 | .3079 | .1158 | .1609 |
| CAE | .1091 | .2008 | .0634 | .1526 | .2305 | .1192 | .1200 | .1038 | .0630 |
| Best baseline, per cell | .6326 | .4267 | .3853 | .7202 | .7540 | .5913 | .5505 | .6834 | .6348 |
| **Our margin** | **+.2082** | **+.3261** | **+.2534** | **+.1366** | **+.0410** | **+.2257** | **+.1461** | **+.1356** | **+.1277** |

Table 1: All models trained on the same $10{,}000$ CLEVR images for $10$ epochs and evaluated on $320$ held-out images per dataset by the same function. Tetrominoes and Multi-dSprites are zero-shot. ARI / F-IOU / O-IOU are foreground ARI, foreground IoU and matched-object IoU on the $64 \times 64$ patch grid. Every model, ours included, has the minimum-region-size rule of Section 3.1 applied at the single value $f = 0.01$ fixed on CLEVR validation. It leaves the baseline rows unchanged to three decimals except AKOrN (conv) on CLEVR foreground IoU, which improves from $.4219$ to $.4267$; the penultimate row uses the better of the two for every baseline cell we measured. That row takes the best baseline *in each cell separately*, which no single baseline attains; our margin is measured against that composite. Single training seed throughout. All models train on CLEVR only, but the conditions are not symmetric: the baselines' cluster count and background mapping are selected on each target dataset's own validation split, while our one free scalar is fixed on CLEVR validation and transferred unchanged (Appendix E).

| $T$ | FG-ARI | FG-IOU | OBJ-IOU | $\Delta$ FG-ARI |
|---|---|---|---|---|
| 1024 | .8421 | .7545 | .6410 | — |
| 512 | .8402 | **.7594** | .6340 | $-.0019$ |
| 256 | .8159 | .7358 | .6084 | $-.0262$ |
| 128 | .5796 | .4988 | .3985 | $-.2625$ |

Table 2: CLEVR, model fixed, readout window $T/2$. Halving the rollout is free; a quarter costs $0.02$–$0.03$; an eighth fails. This table and the scaling figures below are reported *without* the minimum-region-size rule, because both are within-model comparisons at a fixed readout and the rule would shift every row by the same small amount.

| | FG-ARI | FG-IOU | OBJ-IOU | regions |
|---|---|---|---|---|
| control | .8648 | .8028 | .6925 | 7.8 |
| per-patch shift | .0735 | .7358 | .0325 | 562.3 |
| gates, not spikes | .8648 | .8028 | .6925 | 7.8 |

Table 3: First $40$ CLEVR reserved images. Destroying cross-patch timing collapses the partition (FG-ARI, OBJ-IOU) while barely moving foreground IoU. The whole readout, foreground decision included, is recomputed from the shifted traces; foreground IoU survives because it compares only the *union* of predicted non-background patches against the true foreground and is invariant to how that union is partitioned, and shattering one region into $562$ leaves the union nearly unchanged. Substituting the phase gates for the spikes changes nothing at all.

$0.891 \pm 0.111$ for pairs inside one object, $0.047 \pm 0.048$ for pairs in different objects, and $0.099 \pm 0.106$ for object–background pairs. The within-object value is nineteen times the across-object value, and the across-object value sits near the floor of the measure.

**Per-patch circular shift.** Each patch's four component traces are rolled by one offset drawn uniformly over the readout window, independently per patch. Rate, waveform and within-patch structure survive; only the alignment between patches is destroyed.

The readout's *rules* are unchanged — the image-conditioned graph, the $Q \geq 0.5$ link threshold, the $8$-patch component floor, the largest-component background heuristic and the greedy merge — but every quantity they operate on is recomputed from the shifted traces. In particular the foreground is *not* held fixed and is *not* rate-based: the readout passes every patch to the grouping step as a candidate and takes the largest surviving synchrony component as background, so the foreground decision is itself a function of the shifted affinity. The ablation therefore re-runs the whole readout and differs from the control in one input only.

**Gates in place of spikes.** The third row replaces `component_spikes` with `component_gates` and changes nothing else. Every mask is reproduced exactly, on all $40$ images and to every digit of all three metrics. The reason is visible in the traces: with $s_{ia} = g_i e_{ia}$, the firing indicator $e$ equals $1$ in $99.992\%$ of settled steps, the correlation between $s$ and $g$ is $0.9997$, and the $0.008\%$ of steps where the neuron is silent never change a thresholded decision. The timing that the shift ablation shows to be necessary is therefore the timing of the oscillatory gate. The spiking layer carries it without loss, but the readout cannot tell the two apart, and no claim should be made for the threshold

nonlinearity on this evidence.

### 3.6 Limitations

**Object count.** Our readout groups patches better than it counts objects. With the minimum-region-size rule it predicts $5.15$ objects per image on CLEVR against a true $6.36$, matching the exact count on $25.0\%$ of images; $2.64$ against $2.49$ on Multi-dSprites ($53.1\%$); and $5.28$ against exactly $3$ on Tetrominoes ($9.4\%$). Without the rule the same model predicts $8.04$, $6.62$ and $21.07$, matching on $17.5\%$, $28.1\%$ and *none* of the $320$ Tetrominoes images, so most of the count error was single-patch fragments rather than genuine oversegmentation. Letting $f$ be chosen on each dataset's own validation split instead of transferred from CLEVR removes most of the remainder — $f = 0.10$ gives $3.02$ predicted objects on Tetrominoes, $86.9\%$ exact, and raises foreground IoU further to $.8066$ — but that uses target-domain labels, so we report the transferred value as the headline.

Two cautions about this metric. First, Tetrominoes contains exactly three objects in every image, so a model that always answers three is exactly right on every image without grouping anything: the fixed-$k$ baselines all score $100\%$ there, CtCAE included at FG-ARI $0.0756$. The count is only informative where the true count varies, and on Multi-dSprites, where it does, the same baselines score $0\%$ exact against our $53.1\%$. Second, the residual error is a predicted consequence of the objective: the model is trained without a reconstruction term, and a reconstruction loss is what penalises splitting one object into two parts that must then be reconstructed independently. Every baseline in Table 1 optimises such a term; ours does not, which is also what makes it cheap to train.

**Seeds.** Every number reported here is a single training seed. We measured the effect of reseeding one baseline, which moved FG-ARI by up to $0.134$ on CLEVR and $0.009$ on Multi-dSprites. All margins in Table 1 exceed that range, but this is evidence about baseline variance and not an error bar on our own model.

**Protocol.** The patch protocol is not comparable with pixel-protocol numbers reported in the cited works, and the held-out split is reserved from the training corpus rather than an independent benchmark test set. Figures from other papers appear in this section only as each model's comparison with itself (the baseline-fairness comparison), never as entries in Table 1.

## References

Dan, T.; and Wu, G. 2026. From Cortical Synchronous Rhythm to Brain Inspired Learning Mechanism: An Oscillatory Spiking Neural Network with Time-Delayed Coordination. *arXiv preprint arXiv:2605.01656*.